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Project Title : USA Housing Market Analysis & Modeling

1. Project Overview

This project analyzes a dataset of 5,000 homes in the US market to figure out what factors actually drive housing prices. By looking at indicators like regional income, house age, and numbers of rooms, the goal was to find patterns in the housing market and build a model to predict home prices.

The Project Steps:

A) Data Exploration (Correlation Analysis): Created a correlation matrix to see how different variables have a relationship to each other and to find out which factors have the biggest impact on a house's price.

B) Price Prediction (Linear Regression): Built a standard Linear Regression model to predict actual house prices based on features like area income and house age. This helps us see how well we can estimate a home's value using neighborhood data.

C) Grouping Houses (K-Means Clustering): Used K-Means clustering to automatically group similar houses together based on their characteristics, allowing us to see different house tiers.

D) Database Management (MongoDB Compass): Moved the dataset into MongoDB Compass (NoSQL database) to store, sort, and query the data efficiently. Graphed Correlation matrix. Performed K-Means clustering while also evaluating the models performance.

2. Key Findings & Conclusions

A) Income is the strongest factor: The data shows that the average income of a neighborhood is the single most important factor in determining a house's price. It matters significantly more than smaller details like the exact number of bedrooms.

B) Automatic Grouping Works: The K-Means clustering algorithm did a great job of sorting neighborhoods into high-value and baseline markets entirely on its own, proving that machine learning can group real estate tiers without human help.

C) Databases Make Searching Fast: Moving the data into MongoDB made it incredibly easy to manage. Made queries that filter the database to find the top 10 most expensive. top 10 cheapest homes, 4+ bedrooms, over \$2 million, California and under \$1 million.

Summary

This project successfully establishes an end-to-end data pipeline that transforms raw real estate data into actionable regional market intelligence. By combining exploratory data analysis and predictive modeling within an Anaconda notebook environment with the algorithmic workflows of RapidMiner and querying infrastructure of MongoDB Compass. The project demonstrates a modern housing market analysis.

References

USA-Housing-Price-Prediction by PraveenObulreddy

https://github.com/PraveenObulreddy/USA-Housing-Price-Prediction/blob/main/Regression_%2B_Hyperparameter_Tuning_%2B_Cross_Validation_%2B_Sklearn_Pipeline.ipynb

USA-Housing-Price-Prediction by mukulsinghal001

<https://github.com/mukulsinghal001/USA-Housing-Price-Prediction/blob/main/Linear%20Regression%20%2B%20Hyperparameter%20Tuning%20%2B%20Cross%20Validation%20%2B%20Sklearn-Pipeline.ipynb>

USA Housing - Machine Learning(Linear Regression) by Gopal Chettri

<https://www.kaggle.com/code/gopalchettri/usa-housing-machine-learning-linear-regression>